

DATA SCIENCE TRACK

Santander Product Recommendation

Mid-Project Review



PRESENTER

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REVIEW DATE

06 / 10 / 2026

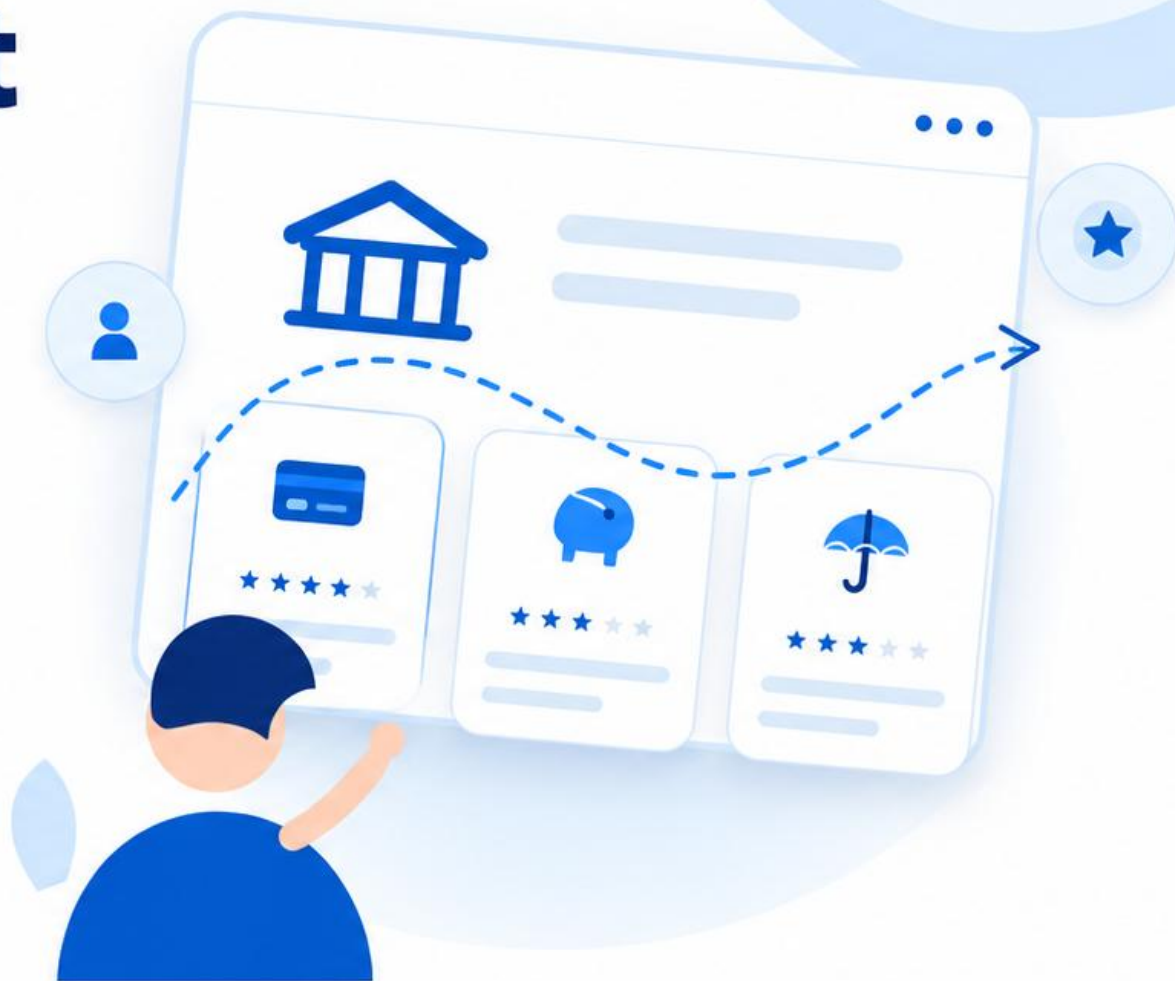


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Santander Product Recommendation dataset



SOURCE

Santander Bank (Spanish)



13.65M

records



956K

users

MONTHLY CUSTOMER SNAPSHOTS



PREDICTION OBJECTIVE



DATASET OVERVIEW



Key
(ncodpers, fecha_dato)



Customer
demographics



Product
ownership

Data overview, EDA, Feature engineering

EDA insight / finding		Data handling / feature response	
1	 Product ownership and acquisition are highly imbalanced. Some products are very common (e.g. ind_cco_fin_ult1), while many others have low acquisition rates.	→	 Consider it in model selection. Use per-product modeling, candidate masking, weighting / sampling, and per-product evaluation.
2	 Customer records are not always continuous across months. There are gaps in customer-month records.	→	 Use as a signal for customer churn / end of relationship when building life cycle. Derive record gaps (e.g. months since last record) to identify inactive or churned customers.
3	 Missingness is structured, especially in income and KYC-like fields. Some fields have a high and non-random missing rate.	→	 Keep missing as a special category or impute using past filling. Preserve informative missingness, or apply past-only forward filling where appropriate, and add missing indicators.
4	 Categorical values are inconsistent. e.g. indrel_1mes has mixed formats: 1, 1.0; 2, 2.0; ...	→	 Normalize categorical values. Convert to consistent format (e.g. 1, 2, ...) before encoding and merge overlapping signals.
5	 renta has negative values, heavy missingness, right-skewed distribution, and a large range. Contains invalid / anomalous values (negative) and many missing entries.	→	 Clean and transform renta. Set negative values to null, impute nulls using regional median then global median, and apply log transform (income_log).

Modeling formulation



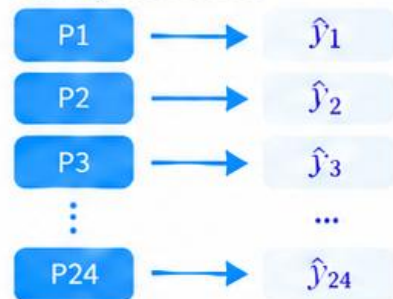
For product p : $y_p = 1$ if state changes 0 at $t - 1$ to 1 at t ; otherwise $y_p = 0$.

1 24 separate product models

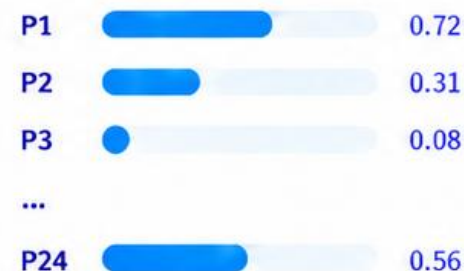
Input features



24 product-specific binary classifiers



24 probabilities



sort + mask owned products $\rightarrow$ top 7



2 One joint model

Input features



Expand into 24 candidate-product samples

(product_id = P1)

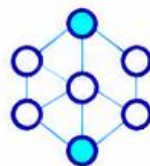
(product_id = P2)

(product_id = P3)

...

(product_id = P24)

Joint binary model



1 probability per expanded sample
 $\rightarrow$ 24 probabilities



sort + mask owned products $\rightarrow$ top 7



Experiment Results



model	config	result	
popularity - based		0.01883	
24-lightgbm models	no FE	0.02404	
24-lightgbm models	sample selection - no FE	0.02518	
1 joint lightgbm model	sample selection - no FE	0.02510	
24-lightgbm models	sample selection + FE	0.02666	
1 joint lightgbm model	Sample selection + FE	0.02714	Best



Results are reported on the test set.

Best result: 0.02714 with 1 joint lightgbm model.

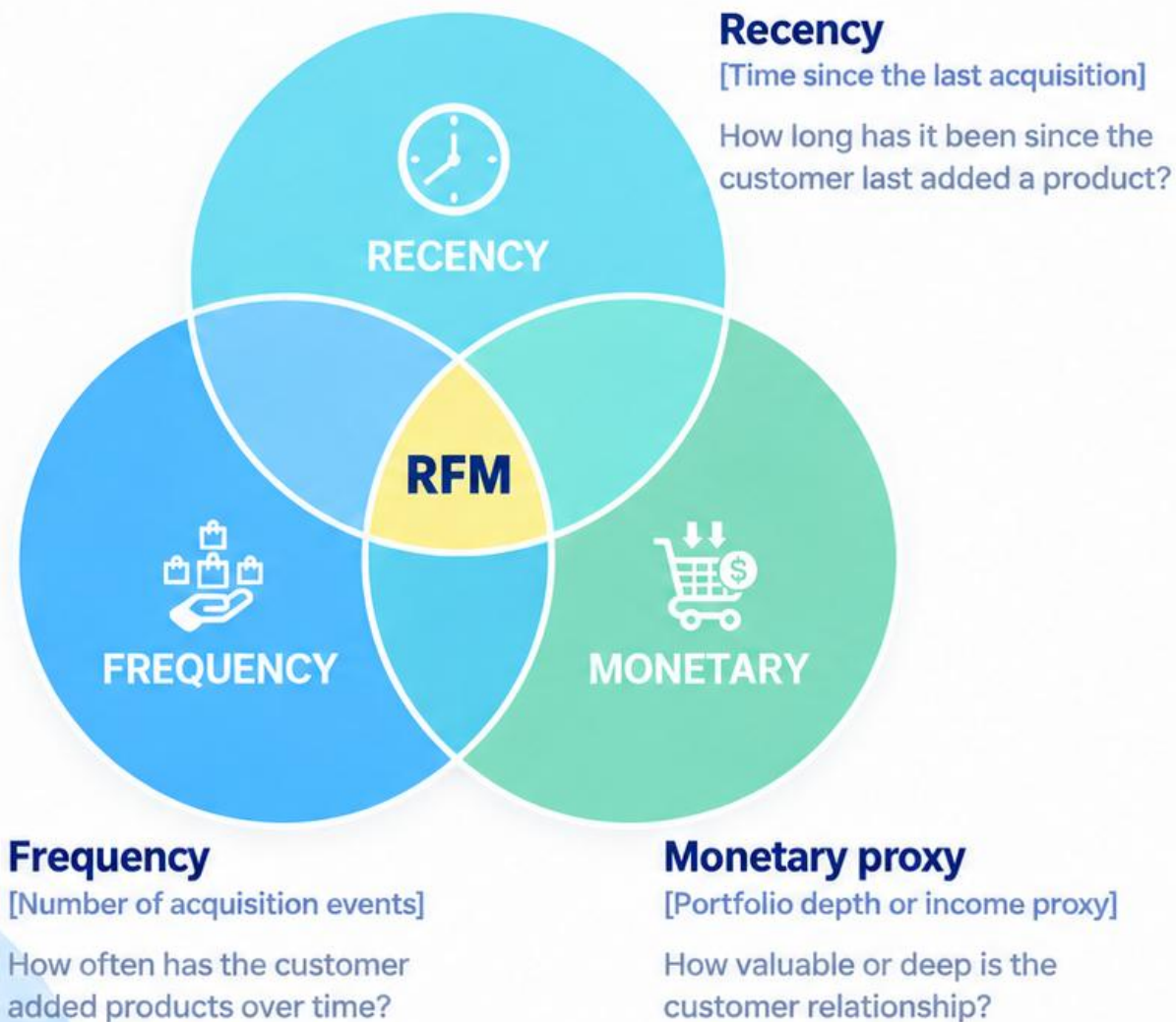
End-to-end data pipeline



Out-of-core processing

Use Parquet + DuckDB so EDA and feature engineering can query the full dataset without loading everything into RAM.

RFM proxy from product events



Tiering experiments and current result



Tiering methods tested

- Quantile tiering
- 1D K-Means
- Rule-based tiers



K candidates

K = 3 K = 4 K = 5



Monetary proxies tested

- Portfolio depth
- Income proxy (renta)



Selection criteria

- behavioral separation
- temporal stability
- interpretability



Current selected result

R → KMeans K=4

F → KMeans K=4

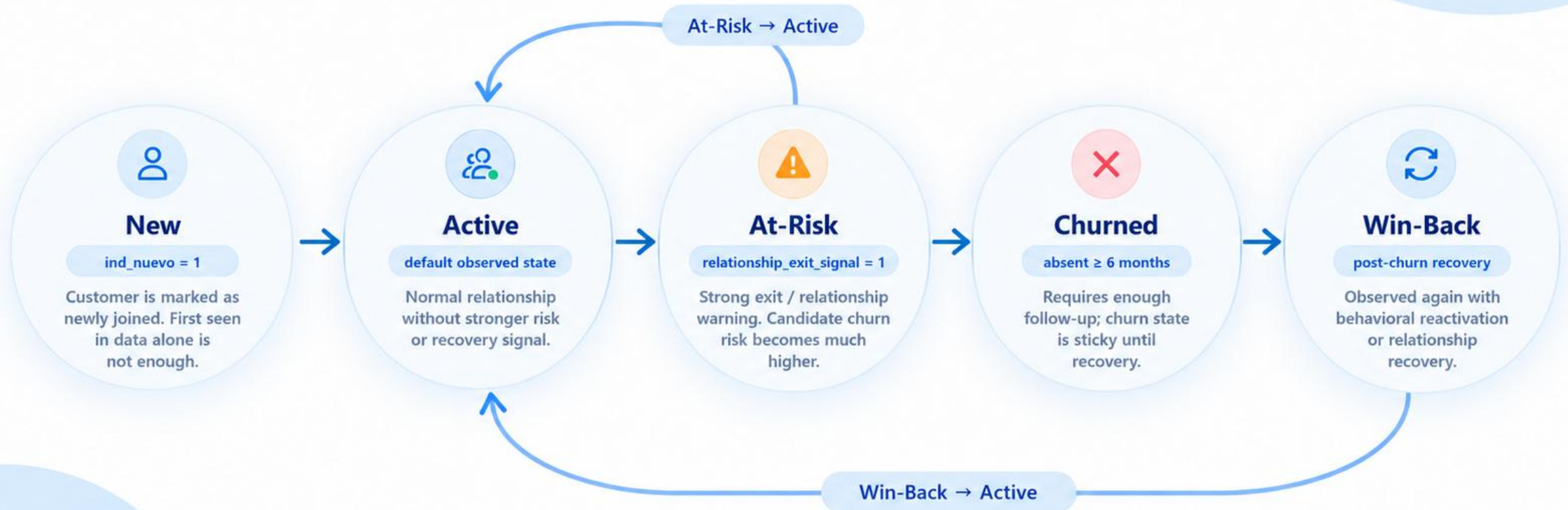
M → Rule K=5



NEVER / MISSING kept as special groups.

Lifecycle states and assignment rules

Lifecycle is modeled as transitions between behavioral states, not a one-way funnel.



Objective

Identify each customer's current relationship state and track transitions over time to support retention, churn monitoring, and re-activation analysis.



How to select churn signal

Identify customers who are likely to churn (become inactive)

Candidate signals

	Absence gap Months since last observed record for the customer	SELECTED Clear and strong relationship with return behavior
	tiprel_1mes = 'P' (former) Customer is no longer primary in the last month	NOT SELECTED Not terminal enough (97.53% become active next month)
	Inactive flag Customer marked as inactive in the dataset	NOT SELECTED Still not a true exit signal (20,601 acquisitions while inactive)
	Portfolio closure All products closed	NOT SELECTED Still reversible (25.76% reacquire next month)

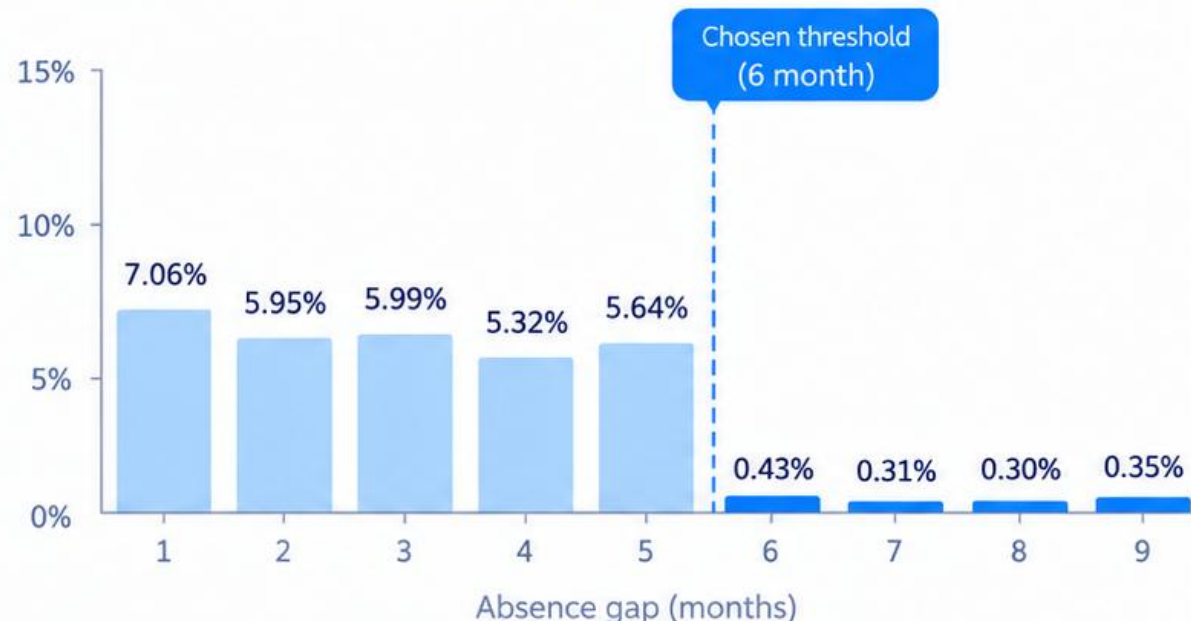


Selected at-risk signal

Absence gap ≥ 6 months

Customer return rate by absence gap

Share of customers who reappear (have a record) in the next month



How to select at-risk signal

Identify customers who are likely to churn (become inactive)

3-month churn rate by candidate signals

Signal		3-month churn rate	Support
All customers (baseline)		0.455%	5,302,312
Inactive		0.220%	2,609,320
Inactive $\geq$ 2 months		0.243%	1,745,901
Recent product drop		0.972%	175,034
Close all products		5.523%	26,405
Relationship exit		46.15%	4,195



Relationship exit signal (selected)

relationship_exit_signal = 1 if any of the following holds:

- $\text{tiprel_1mes} = P$ (former)
- $\text{indrel} = 99$
- $\text{indrel_1mes} \in \{3, 4\}$
- ult_fec_cli_1t is observed



Why this signal?

- ✓ Highest 3-month churn rate (46.15% vs 0.455% baseline).
- ✓ Good support (4,195 customers).
- ✓ Interpretable and easy to compute.

CLV Proxy

Objective: Estimate future customer product activity / value from historical acquisition behavior.

Proxy: no true transaction revenue / profit.

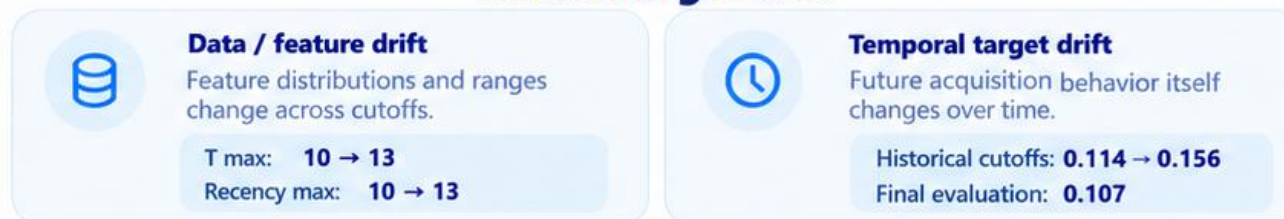
Probabilistic CLV



ML CLV



Remaining issues



Summary

What I have done



EDA



Feature engineering



Built the data pipeline



Modeling



Created features to understand customer personas

What I learned



Processing and reading large files with Parquet



RFM methods



CLV methods



Customer lifecycle tiering



EDA workflow: insight → data handling decisions

Thanks for listening

Questions, feedback, and suggestions are welcome.

